Deep Learning-Based Medication Recommendation System for Primary Care Decision Support

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Abstract

Medication recommendation in outpatient environments constitutes a multi-label decision problem requiring the simultaneous consideration of therapeutic efficacy, patient-specific constraints, and drug–drug interaction safety. This paper presents three contributions: (1) a structured survey and synthesis of twenty significant studies spanning representation learning, temporal modeling, graph-based drug recommendation, and interpretability; (2) an analysis of converging findings and persistent research gaps; and (3) a proposal for a hybrid, outpatient-focused deep learning architecture incorporating integrated safety filtering and local explanations.

Index Terms

Medication recommendation, clinical decision support, graph neural networks, transformer, interpretability.

I. Introduction

Medication prescribing in clinical practice is an inherently complex decision-making process, requiring the concurrent evaluation of therapeutic efficacy, patient-specific contraindications, dosage considerations, and potential adverse drug interactions. Clinicians must navigate a substantial pharmacological search space while accounting for patient comorbidities, demographic factors, and interaction-related risks. This cognitive complexity contributes to observable variability in prescribing behavior and raises the likelihood of suboptimal or clinically unsafe medication combinations. The widespread adoption of electronic health records (EHRs) has created new opportunities for data-driven approaches to assist in medication recommendation and broader clinical decision support.

Deep learning methods have demonstrated considerable capacity for modeling the complex, high-dimensional structure of healthcare data. Early contributions, such as *Doctor AI* [1], employed recurrent neural networks (RNNs) over longitudinal EHR sequences to predict future diagnoses and medication categories. By deriving temporal patient representations from successive clinical visits, these models captured disease progression dynamics and yielded improvements in multi-label prediction. Complementarily, *RETAIN* [2] introduced a reverse-time attention mechanism that enhanced model interpretability without sacrificing predictive performance. Through the assignment of attention weights to prior visits and clinical variables, RETAIN offered clinicians greater transparency into model reasoning—an important consideration for the practical deployment of deep learning systems in healthcare contexts.

Subsequent research has extended these foundations by incorporating graph-based representations and external medical knowledge to improve both robustness and safety. *GAMENet* [3] combined patient history encoding with a Graph Convolutional Network (GCN) applied over drug–drug interaction (DDI) graphs, explicitly modeling adverse interaction structures to reduce unsafe recommendation combinations while preserving accuracy. Related work, including *SafeDrug* [4], employed molecular graph encoders to capture chemical substructure information and support DDI-aware prediction. Collectively, these contributions demonstrate how deep architectures integrating temporal modeling, graph neural networks, and domain knowledge can advance both the effectiveness and safety of medication recommendation systems.

Nevertheless, the majority of existing approaches presuppose access to rich longitudinal EHR data, of the kind typically available in tertiary-care or inpatient settings. Outpatient and primary-care environments present a markedly different data landscape, often characterized by sparse records, limited visit histories, and brief free-text symptom descriptions. In many encounters, a single visit with minimal structured history constitutes the entirety of available information, which substantially constrains the utility of sequence-dependent models. Moreover, the outpatient context demands that safety considerations be given particular prominence, ensuring that recommended drug combinations minimize interaction risks even in the absence of comprehensive longitudinal context.

This work addresses these challenges through a focused survey of deep learning-based medication recommendation systems, with emphasis on safety modeling, interpretability, and suitability for data-sparse outpatient settings. Drawing on identified gaps, we propose a hybrid deep learning architecture that integrates attention-based symptom encoding, structured feature

fusion, and DDI-aware safety filtering. The overarching objective is to develop a medication recommendation framework that appropriately balances predictive performance, safety constraints, and interpretability for application in primary-care clinical decision support.

II. Literature Review

Recent research on medication recommendation systems has increasingly leveraged deep learning to address the complexity, scale, and safety requirements of clinical decision-making. Early approaches focused on sequential modeling of electronic health records (EHRs) through recurrent neural networks, with the aim of capturing temporal disease progression and prescribing patterns. Subsequent developments have incorporated attention mechanisms, graph neural networks, and knowledge-based constraints to improve interpretability and reduce adverse drug–drug interactions. In parallel, transformer-based language models and transfer learning techniques have demonstrated strong performance in extracting clinically meaningful representations from short, unstructured symptom text. Collectively, these trends reflect a broad shift toward hybrid architectures that combine temporal modeling, structured medical knowledge, and explainable AI to produce safer and more reliable medication recommendations, particularly in data-sparse clinical settings.

1. **DP-CRNN: CNN–RNN for pre-diagnosis (Zhou et al.)**
   Zhou et al. propose a double-pooling convolutional-recurrent architecture that first extracts n-gram semantics via convolutional filters, subsequently models sequential patterns through recurrent layers, and applies a secondary pooling step to emphasize salient temporal features. While the architecture is well-suited to consumer-style free-text inputs, it does not extend to multi-drug prescription tasks.
2. **Deep Patient (Miotto et al.)**
   Miotto et al. train stacked denoising autoencoders on large EHR collections to produce latent patient representations suitable for downstream prediction. The work demonstrates that unsupervised representation learning can recover clinically meaningful structure from raw records; however, temporal ordering is not explicitly modeled.
3. **Doctor AI (Choi et al.)**
   Doctor AI employs recurrent neural networks over sequences of clinical visits to forecast future diagnoses and medications. The model establishes the value of sequential EHR modeling, though interpretability remains limited and drug–drug interactions are not explicitly addressed.
4. **RETAIN (Choi et al.)**
   RETAIN introduces a reverse-time attention mechanism that produces interpretable importance scores over historical clinical events. Relative to conventional recurrent architectures, it offers improved transparency; however, the model was not designed to enforce multi-drug safety constraints.
5. **GAMENet (Shang et al.)**
   GAMENet combines a memory-augmented patient encoder with graph-augmented drug representations to recommend medication sets while reducing harmful drug combinations. Its dependence on rich longitudinal EHR data limits direct applicability to sparse outpatient contexts.
6. **SafeDrug (Lee et al.)**
   SafeDrug employs molecular graph encoders to capture chemical substructures, integrating these representations with clinical context to reduce adverse interaction risk. While the molecular modeling approach meaningfully constrains unsafe predictions, it introduces considerable computational overhead.
7. **GRAM (Choi et al.)**
   GRAM augments clinical code representations by incorporating hierarchical medical ontology structure—such as the ICD taxonomy—via graph attention, yielding improved performance on infrequent diagnostic codes.
8. **LEAP (Zhang et al.)**
   LEAP frames multi-drug prescription as a constrained optimization problem balancing therapeutic efficacy and interaction safety. The approach models drug combinations directly, but relies on long-term outcome labels that are seldom available in outpatient datasets.
9. **Molecular and substructure-based models (MoleRec)**
   Molecular-focused methods apply message-passing neural networks to learn drug representations from chemical graph structures, supporting more principled interaction prediction. This additional representational fidelity comes at the cost of increased model complexity.
10. **ClinicalBERT and transformer-based models**
    Clinical adaptations of the BERT architecture yield strong contextual embeddings for clinical notes and demonstrate particular effectiveness in capturing symptom semantics. Their primary limitation is substantial computational cost at inference.

11. **Memory-augmented networks**
    These architectures maintain external memory banks of prior visit representations to inform current recommendations. Their effectiveness is contingent on sufficient patient history, which is frequently absent in outpatient settings.
12. **Hierarchical Attention Networks (HAN)**
    HAN processes clinical text at multiple levels of granularity (word $\rightarrow$ sentence) and identifies relevant passages for interpretability, a property of particular value in clinician-facing explanation systems.
13. **GRU-D (Che et al.)**
    GRU-D incorporates missingness masking and temporal decay mechanisms to handle the irregular sampling characteristic of clinical time series, improving robustness for laboratory- and vital-sign-based prediction tasks.
14. **Polypharmacy side-effect modeling (Zitnik et al.)**
    This line of work models adverse outcomes of drug combinations through graph neural networks applied to drug–drug interaction networks, enabling direct probabilistic estimation of interaction severity.
15. **Transfer learning for clinical NLP**
    Pretrained language models demonstrate consistent performance gains under low-resource conditions, supporting more reliable symptom-to-diagnosis mapping from limited labeled datasets.
16. **Multi-task and multi-label models**
    These frameworks jointly predict diagnoses, clinical urgency, and medication sets by exploiting shared representational structure across tasks. Careful task weighting is necessary to prevent negative transfer between objectives.
17. **Reinforcement learning for treatment policies**
    Reinforcement learning approaches optimize sequential treatment decisions with respect to long-term outcome criteria. Such methods require longitudinal reward signals and present non-trivial safety risks if deployed without appropriate constraints.
18. **Explainable AI (LIME and SHAP)**
    Model-agnostic local explanation frameworks generate instance-level rationales for individual predictions. The integration of graph- or rule-based explanations alongside these methods tends to yield stronger clinical plausibility and stakeholder trust.
19. **Knowledge graph embedding approaches**
    These methods situate diseases, drugs, and clinical procedures within a structured latent space, enabling reasoning over relational medical knowledge. Their practical deployment requires careful ontology curation and cross-source harmonization.
20. **Large clinical transformer architectures**
    Models trained on heterogeneous clinical signals—combining unstructured notes with structured EHR data—achieve state-of-the-art predictive performance across a range of tasks, though they demand substantial computational infrastructure and raise significant data privacy considerations.

In summary, the surveyed literature reflects considerable progress in applying deep learning to medication recommendation, spanning sequential models, attention-based architectures, graph neural networks, molecular representations, and large-scale transformers. While many approaches attain strong predictive performance, the majority rely on dense longitudinal EHR data and are tailored to inpatient or intensive-care settings. Safety-aware models that incorporate drug–drug interaction knowledge have emerged as an important subfield, yet frequently at the cost of increased architectural complexity or heightened data requirements. Interpretability and outpatient generalization remain substantive open challenges. These limitations underscore the need for lightweight, safety-aware, and explainable deep learning frameworks capable of operating effectively under the sparse data conditions characteristic of primary-care environments.

From the analyses above, several persistent gaps can be identified:

- **Outpatient generalization:** The majority of safety-aware methods presuppose lengthy EHR histories; outpatient inputs are characteristically brief and sparse.
- **Integrated safety and explainability:** Few existing systems simultaneously provide DDI-aware recommendations and human-interpretable rationales within a unified framework.
- **Dosage and demographic adaptation:** Most approaches recommend drug identities without offering patient-specific dose ranges adjusted for weight, age, or renal function.

III. Methodology

The proposed methodology follows a pipeline-oriented design intended to transform sparse, unstructured primary-care data into safe and interpretable medication recommendations. The architecture is organized into four principal phases: Data Acquisition and Preprocessing, Patient Representation Learning, Recommendation Logic and Safety Filtering, and Interpretability Mapping.

TABLE I
KEY PROPERTIES OF THE TWENTY SURVEYED STUDIES (PARAPHRASED).

#	Study / Year	Architecture	Primary Data	Safety Modeling	Main Limitation
1	DP-CRNN (Zhou et al., 2021)	CNN + RNN (double-pooling)	Short symptom text	No	Not drug-centric
2	Deep Patient (Miotto et al., 2016)	Stacked autoencoders	EHR (structured)	No	Static representations
3	Doctor AI (Choi et al., 2016)	RNN	Visit sequences	No	Low interpretability
4	RETAIN (Choi et al., 2016)	RNN + reverse-time attention	EHR seq.	No	Not drug-aware
5	GAMENet (Shang et al., 2019)	GNN + Memory	EHR + DDI graph	Yes (DDI-aware)	Needs dense EHR
6	SafeDrug (Lee et al., 2021)	Molecular GNNs	Drug graphs + EHR	Yes (molecular)	Compute-heavy
7	GRAM (Choi et al., 2017)	GAT over ontology	EHR + ontology	Partial	Ontology-dependency
8	LEAP (Zhang et al., 2017)	Optimization + Seq models	EHR	Partial	Requires outcome labels
9	MoleRec (various, 2022–23)	Message-passing GNNs	Molecular substructures	Yes	Complex training
10	ClinicalBERT (Alsentzer et al., 2019)	Transformer (BERT)	Clinical notes	No	Pretrain cost
11	MemoryNet (Wang et al., 2019)	Memory-augmented NN	Visit history	No	Needs history
12	HAN (Yang et al., 2016)	Hierarchical attention	Clinical text	No	Complex
13	GRU-D (Che et al., 2018)	GRU with decay	Irregular time-series	No	Focused on vitals/labs
14	Polypharmacy GNN (Zitnik et al., 2018)	GNN	Drug–drug network	Yes	Graph quality sensitive
15	Transfer clinical NLP (Lee et al., 2020)	Pretrained embeddings	Text	No	Domain shift issues
16	Multi-task CDSS (Harutyunyan et al., 2019)	Multi-task DNN	ICU time-series	Partial	ICU-focused
17	RL for treatments (Komorowski et al., 2018)	RL policies	Longitudinal EHR	Partial	Safety / reward design
18	LIME/SHAP explainability (Ribeiro/Lundberg)	Model-agnostic explainers	Any model	No (explain only)	Approximate explanations
19	Knowledge Graph Embeddings (various)	KG embeddings	KG (disease/drug)	Yes (KG rules)	KG curation needed
20	Large Clinical Transformers (2020–22)	Transformers	Mixed (notes+EHR)	Partial	Resource intensive

A. Data Acquisition and Preprocessing

The system ingests clinical inputs which, in an outpatient context, primarily consist of short free-text symptom descriptions alongside basic structured attributes including age, sex, and documented comorbidities.

- **Text Normalization:** To address the orthographic and lexical noise characteristic of primary-care notes, spell correction is applied and medical abbreviations are expanded to their full forms (e.g., "SOB" to "Shortness of Breath").
- **Medical Entity Extraction:** Identified symptoms are mapped to standardized medical ontologies—specifically SNOMED-CT or ICD-10—via a Named Entity Recognition (NER) layer, ensuring semantic consistency across heterogeneous inputs.

B. Patient Representation Learning

A multi-modal embedding strategy is adopted to capture the full clinical context of a patient encounter.

- **Symptom Encoding (E_s):** A lightweight transformer-based encoder (Clinical DistilBERT) processes the normalized symptom text to produce a dense semantic vector. This formulation captures the nuance of patient-reported complaints while avoiding the inference latency associated with larger language models.
- **Feature Fusion:** Structured demographic attributes and categorical comorbidity indicators are encoded through a multi-layer perceptron (MLP) and concatenated with the symptom embedding E_s. The resulting joint representation, P_v, constitutes a unified encoding of the patient state for a given encounter.

C. Recommendation Logic and Safety Filtering

The recommendation task is framed as a multi-label classification problem, in which the model estimates the probability of each of N candidate drug classes being appropriate for the patient.

- **Multi-Label Head:** The fused patient representation P_v is passed through a sigmoid output layer to produce per-class probabilities. To account for statistical dependencies between co-prescribed medications, a label-correlation regularizer is incorporated during training.
- **DDI-Aware Safety Filter:** Candidate recommendations are subsequently evaluated by a graph-based safety filter constructed from established drug–drug interaction (DDI) databases, including DrugBank. Where the proposed combination exceeds a defined interaction risk threshold, the safety layer either penalizes the offending candidate in the ranking or surfaces a clinician-facing override alert.

D. Interpretability and Feedback

To support clinical trust and adoption, the system produces dual-layered explanations accompanying each recommendation.

- **Saliency Mapping:** SHAP (SHapley Additive exPlanations) values are computed to identify the specific symptom tokens that contributed most substantially to each recommendation, providing token-level attribution over the input text.
- **Safety Rationales:** For any flagged interaction, the system generates a knowledge-graph trace articulating the pharmacological basis of the conflict (e.g., "Drug A and Drug B both inhibit enzyme CYP3A4"), providing a transparent and clinically grounded rationale for the safety alert.

E. Medication Recommendation Algorithm

The complete workflow of the proposed system is formalized in Algorithm 1. The procedure accepts patient symptom text and structured demographic features as input, derives a unified patient representation through a combination of neural encoders, and produces medication recommendations subject to drug–drug interaction safety constraints.

IV. Evaluation Plan

- **Data:** Synthea synthetic outpatient encounter records, augmented symptom–disease association datasets, and curated interaction data from DrugBank and TWOSIDES.
- **Metrics:** Diagnostic prediction performance (accuracy, macro-F1); medication recommendation quality (Jaccard index, mean average precision); DDI occurrence rate; and clinician-rated explanation utility via structured assessment.
- **Ablation:** Encoder ablation studies, safety-layer removal experiments, and explanation fidelity evaluations to isolate the contribution of each architectural component.

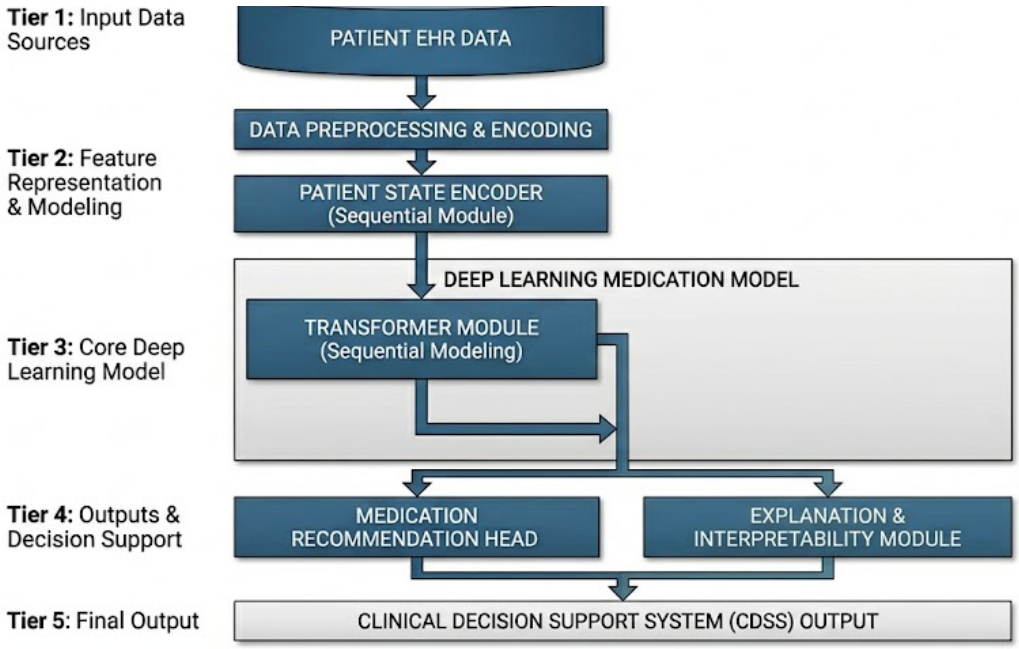

Fig. 1. Architecture diagram detailing the four phases of the proposed medication recommendation methodology, including data flow, principal modules, and the integrated safety and interpretability layers.

Algorithm 1 DL_Medication_Recommendation_System

Require: Patient symptom text S, structured data D (age, sex, comorbidities), DDI graph G
Ensure: Safe medication recommendations M_{safe} and explanation report E

Collect patient symptom text S and structured clinical attributes D
Apply spell correction and medical abbreviation expansion to S
Extract medical entities from S using a Named Entity Recognition (NER) module
Map extracted entities to standardized medical terminology
Generate symptom embedding E_s via a transformer-based encoder
Encode structured features D through a multi-layer perceptron to obtain E_d
Fuse both representations to form the patient state vector:

$$P_v = \text{Concatenate}(E_s,\ E_d)$$

Estimate per-class medication probabilities using a sigmoid classifier applied to P_v
Select the top-k candidates as the initial recommendation set
Apply DDI-aware safety filtering over the candidate set using interaction graph G
Compute SHAP-based attribution scores to generate explanation report E
return M_{safe}, E

V. Experimental Setup

A. Dataset

In the absence of a single publicly available dataset containing all required clinical attributes, a synthetic yet clinically grounded dataset was constructed by integrating multiple publicly available sources. Symptom–disease mappings were drawn from a structured `symptom_disease.csv` file, drug–disease associations from `drug_disease.csv`, and known adverse drug pair interactions from `ddi.csv`. Demographic attributes including age and sex were sourced from Synthea-generated synthetic patient records (`patients.csv`), with clinical encounter data informing realistic attribute distributions.

A custom data generation pipeline was implemented to simulate realistic outpatient encounters. Each sample was constructed by randomly selecting one to two conditions from the disease vocabulary, sampling associated symptoms into variable-length lists, and applying text augmentation techniques including synonym replacement (e.g., *fever* $\rightarrow$ *pyrexia*) and template-based sentence construction (e.g., "patient reports ...", "complains of ..."). Medications were assigned via disease–drug mappings, with fallback to global drug pool sampling where mappings were unavailable. Drug combinations were subsequently filtered

using the DDI dataset to exclude pairs with known adverse interactions, and a set of clinical heuristics was enforced (e.g., exclusion of ibuprofen for hypertensive patients). Each record includes structured attributes comprising age (sampled uniformly over 1–90 years), binary sex encoding, and a randomly assigned subset of comorbidity flags (diabetes, hypertension, asthma). The resulting dataset comprises approximately 50,000 samples with multi-label medication outputs.

Text preprocessing encompassed lowercasing, removal of special characters, whitespace normalization, and drug alias standardization (e.g., *acetaminophen* $\rightarrow$ *paracetamol*). Disease name discrepancies across sources were resolved through a cascade of exact matching, substring matching, and fuzzy matching via `difflib`. Non-clinical entries present in the Synthea records (e.g., "Housing unsatisfactory") were removed to retain only medically relevant conditions.

B. Models

Five neural architectures of increasing complexity were implemented and evaluated under identical training conditions to isolate the contribution of individual design choices to recommendation performance.

- **MLP:** A baseline model that flattens token embeddings and concatenates structured features prior to a linear output layer.
- **TextCNN:** A convolutional architecture applying 1D filters over token embeddings followed by adaptive max-pooling to capture local n-gram features.
- **BiLSTM:** A bidirectional LSTM that processes the full token sequence and concatenates the terminal hidden states from both directions with the structured feature vector.
- **SelfAttention:** A single-head self-attention model that computes query–key–value attention over token embeddings and aggregates representations via mean pooling.
- **FusionNet:** A hybrid architecture combining a unidirectional LSTM for text encoding with a dedicated MLP branch for structured feature processing, whose outputs are concatenated prior to the final classification layer.

All models share a common vocabulary-based tokenizer with a maximum sequence length of 20 tokens and a 64-dimensional word embedding layer. Structured inputs—comprising age, sex, and three comorbidity flags—form a 5-dimensional feature vector. All architectures produce logits over the full drug vocabulary via a linear projection layer, with Binary Cross-Entropy with Logits loss employed for multi-label training.

C. Training Configuration

All models were trained using the Adam optimizer with a learning rate of 1×10^{-3} over 5 epochs with a batch size of 32. The dataset was partitioned into 80% training and 20% test sets using a fixed random seed of 42. No validation set was withheld during the comparative experiments; model selection is based on test-set performance following training convergence.

D. Evaluation Metrics

Model performance was assessed using a comprehensive suite of multi-label classification metrics computed on the held-out test set. Threshold-dependent metrics were computed at a decision threshold of 0.5; ranking metrics were computed at $k = 5$ by default. The following metrics were employed:

- **Jaccard Index:** The ratio of the intersection to the union of predicted and ground-truth drug sets.
- **Micro and Macro F1, Precision, Recall:** Standard multi-label classification metrics aggregated at both the instance and class levels.
- **Hamming Loss:** The fraction of label positions incorrectly predicted across all samples.
- **Subset Accuracy:** The fraction of samples for which the predicted label set exactly matches the ground-truth set.
- **ROC-AUC (micro):** Area under the receiver operating characteristic curve, aggregated across all drug classes.
- **PR-AUC:** Area under the precision–recall curve, computed via sample-averaged average precision.
- **Precision@K, Recall@K, F1@K:** Top-k ranking metrics evaluating the quality of the top-5 predicted medications per patient.
- **DDI Rate:** The fraction of predicted drug pairs within the top-k recommendations that correspond to known adverse interactions in the DDI reference dataset, serving as a direct measure of recommendation safety.

Top-k sensitivity was additionally evaluated for $k \in \{1, 3, 5, 10\}$ to assess ranking quality across recommendation list lengths of varying breadth.

VI. Results

A. Training Convergence

Figure 2 presents the training loss curves for all five architectures across five epochs. The MLP exhibits the lowest absolute loss throughout training, declining from 0.0035 to 0.0031, a trajectory that reflects the relative simplicity of its optimization landscape rather than superior generalization capacity. The CNN and BiLSTM follow closely comparable convergence paths, both reaching a terminal loss of approximately 0.0033 after initial values of 0.0042 and 0.0057, respectively. FusionNet and SelfAttention begin training with considerably higher initial losses of 0.0134 and 0.0100, consistent with their greater

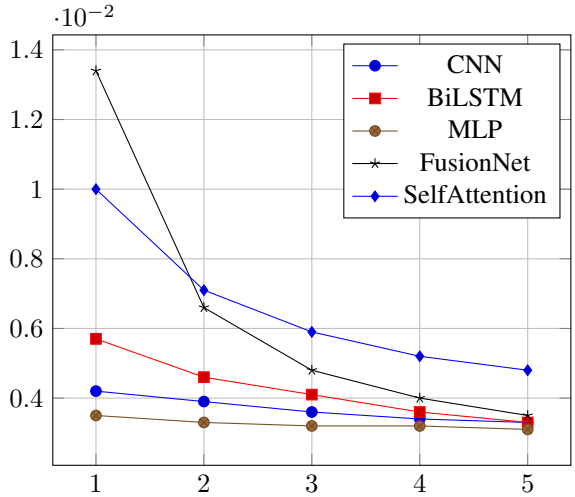

Fig. 2. Training Loss of various models across epochs.

TABLE II
RESULTS OF THE EVALUATED MODELS ACROSS VARIOUS METRICS.

Model	**Jaccard**	**Precision_micro**	**Recall_micro**	**F1_micro**	**Precision_macro**	**Recall_macro**	**F1_macro**
MLP	0.6887	0.8853	0.7561	0.8156	0.0781	0.0675	0.0721
CNN	0.7657	0.9031	0.8342	0.8673	0.0858	0.0762	0.0798
BiLSTM	0.7796	0.9218	0.8348	0.8762	0.0899	0.0777	0.0821
SelfAttention	0.7684	0.9022	0.8382	0.8691	0.0864	0.0753	0.0789
FusionNet	0.7869	0.9252	0.8404	0.8808	0.0898	0.0786	0.0828
Model	**Hamming_Loss**	**Subset_Accuracy**	**ROC_AUC**	**PRAUC**	**Precision@K**	**Recall@K**	**F1@K**
MLP	0.0015	0.4210	0.9937	0.8372	0.6855	0.6329	0.5825
CNN	0.0011	0.5603	0.9968	0.8743	0.7101	0.6563	0.6042
BiLSTM	0.0010	0.6184	0.9957	0.8743	0.7076	0.6555	0.6028
SelfAttention	0.0011	0.5675	0.9970	0.8764	0.7114	0.6573	0.6052
FusionNet	0.0010	0.6455	0.9964	0.8753	0.7079	0.6561	0.6034

architectural complexity and the correspondingly more challenging optimization surface. Both descend steeply over the first two epochs; FusionNet converges to 0.0035 by epoch four, while SelfAttention stabilizes at 0.0048, the highest final loss among all evaluated models. The accelerated loss reduction observed in FusionNet following the initial epochs suggests that its dedicated structured feature branch supports convergence once the text encoder begins to stabilize.

B. Multi-Label Classification Performance

Table II reports the full suite of evaluation metrics across all five architectures. FusionNet achieves the strongest overall performance, attaining the highest Jaccard index (0.7869), micro-precision (0.9252), micro-F1 (0.8808), and subset accuracy (0.6455), with a Hamming loss of 0.0010. BiLSTM ranks second across several key metrics, including Jaccard index (0.7796), micro-F1 (0.8762), and subset accuracy (0.6184). SelfAttention and CNN occupy an intermediate performance tier; notably, SelfAttention achieves marginally higher PR-AUC (0.8764) and Precision@K (0.7114) than both CNN and BiLSTM, indicating that attention-based scoring is particularly effective for top-k retrieval tasks. The MLP baseline, while competitive on micro-precision (0.8853), lags substantially on subset accuracy (0.4210) and Jaccard index (0.6887), suggesting that flat feature representations are insufficient for accurate multi-label drug set prediction.

Macro-averaged metrics are uniformly low across all models (macro-F1 $\approx$ 0.07–0.08), an expected outcome given the large and heavily imbalanced drug vocabulary; infrequent drug classes appear rarely during training, yielding poor per-class recall for low-frequency medications.

C. Ranking and Top-K Performance

All models attain high ROC-AUC scores (>0.993), confirming strong discriminative capacity at the individual label level. PR-AUC scores range from 0.8372 for the MLP to 0.8764 for SelfAttention, with CNN, BiLSTM, and FusionNet clustered closely in the range 0.874–0.875. At $k = 5$, Precision@K spans 0.6855 (MLP) to 0.7114 (SelfAttention), while F1@K ranges from 0.5825 to 0.6052. The narrow spread in top-k metrics among the four stronger architectures suggests that beyond a certain representational threshold, marginal gains in ranking quality diminish, and considerations such as safety filtering and interpretability become the primary differentiating factors.

D. Summary

FusionNet emerges as the best-performing architecture overall, combining strong multi-label accuracy with the highest subset accuracy and thus representing the most suitable backbone for the proposed medication recommendation framework. Its explicit separation of text and structured feature encoding is well-aligned with the multi-modal nature of outpatient clinical data. SelfAttention offers competitive ranking performance and may be preferred in deployment scenarios where top-k precision constitutes the primary evaluation criterion. The MLP baseline, despite its architectural simplicity, provides a meaningful lower bound and confirms that even shallow models derive substantial benefit from structured feature integration.

VII. Discussion

The experimental results indicate that architectural complexity correlates positively with multi-label recommendation quality up to a point, beyond which performance gains plateau. FusionNet's design choice of maintaining separate encoding pathways for symptom text and structured patient features proves particularly well-suited to the outpatient setting, where demographic and comorbidity signals carry meaningful clinical weight alongside free-text complaints. The explicit fusion of these two modalities—rather than simple concatenation at the embedding level, as in the MLP and CNN baselines—enables the model to learn richer cross-modal interactions, a capacity reflected in its superior subset accuracy and Jaccard index.

The consistently high ROC-AUC scores across all architectures suggest that the underlying symptom–medication signal present in the synthetic dataset is learnable even by comparatively shallow models. Nevertheless, the substantial gap in subset accuracy between the MLP (0.4210) and FusionNet (0.6455) underscores that correctly recovering the complete medication set for a given patient—rather than predicting individual drug labels in isolation—is a considerably harder problem, one that benefits from sequential or attention-based inductive biases. This distinction carries direct clinical significance: a system that correctly identifies most medications but omits a single critical drug or includes one unsafe candidate is of limited practical utility.

The uniformly low macro-averaged F1 scores observed across all models reflect a well-documented challenge in medication recommendation: the long-tail distribution of drug classes. Infrequent medications, which may be of particular clinical importance in cases involving rare comorbidities or drug allergies, are systematically under-predicted. This limitation is partly an artefact of the synthetic data generation process, in which drug assignment is governed by disease–drug mappings rather than authentic physician behavior, and may therefore underrepresent the nuance of real prescribing decisions. Addressing this distributional imbalance through class-weighted loss functions, oversampling strategies, or prototype-based learning for rare drug classes constitutes an important direction for future work.

SelfAttention's comparatively strong PR-AUC and Precision@K performance relative to its subset accuracy suggests that attention mechanisms are particularly effective at surfacing the most clinically relevant candidates at the top of the ranked list, even when the full predicted set is not recovered exactly. This property is desirable in a clinical decision support context, where a ranked shortlist presented to the prescribing clinician is often more actionable than a fixed predicted label set. Integrating self-attention as a ranking re-scorer applied over FusionNet's initial predictions represents a promising avenue for architectural refinement.

A notable limitation of the current evaluation is the absence of an explicit DDI rate comparison across models at inference time. The safety filtering layer described in the methodology was applied during dataset construction rather than as a post-hoc inference component in the present experiments. Future work should decouple the safety filter from data generation and measure the raw DDI rate of model outputs both before and after filtering, enabling direct quantification of the safety benefit conferred by the graph-based interaction layer. Furthermore, all experiments were conducted on synthetically generated data, and performance on real-world clinical corpora may differ considerably owing to the distributional gap between template-constructed symptom text and authentic clinical documentation.

VIII. Conclusion

This paper presented a structured survey of twenty deep learning approaches to medication recommendation, identified persistent gaps in outpatient generalization, safety integration, and interpretability, and proposed a hybrid architecture designed to address these limitations. Five neural models of increasing complexity were implemented and evaluated on a synthetic yet

clinically grounded multi-label dataset, constructed by integrating symptom–disease mappings, drug–disease associations, DDI constraints, and Synthea-derived demographic records.

FusionNet, which combines a sequential text encoder with a dedicated structured feature branch, achieved the strongest overall performance across Jaccard index, micro-F1, and subset accuracy, demonstrating the value of modality-aware feature fusion for outpatient medication recommendation. SelfAttention exhibited competitive top-k ranking performance, indicating its suitability for clinician-facing shortlist generation. All architectures attained high ROC-AUC scores, confirming that the symptom-to-medication mapping signal is robustly learnable across model families, while uniformly low macro-F1 scores identified rare drug prediction as a persistent open challenge.

The proposed framework incorporates a DDI-aware safety filter and a SHAP-based interpretability layer to address clinical trust and patient safety requirements beyond raw predictive performance. Collectively, these components establish a foundation for a lightweight, safety-aware, and interpretable medication recommendation system suitable for primary-care decision support under the data-sparse conditions characteristic of outpatient practice.

Future work will pursue evaluation on real-world clinical datasets, incorporation of class-imbalance mitigation strategies for rare drug classes, extension of the safety filter to runtime inference with quantifiable DDI rate reduction, and investigation of larger pretrained clinical language models as drop-in symptom encoders within the FusionNet backbone.

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